

ENGINEERING MATHEMATICS

Unit I — Matrices | Complete Formula & Revision Sheet

⚡ One-Night Revision Weapon · Formula Bible · Exam Ready

📐 Rank · Echelon · Normal Form

💻 Gauss Methods · Iterative Methods

📖 Rouché–Capelli Theorem

📇 25 Flash Cards Included

📋 Table of Contents

1. Course Objectives	Pg 1	2. Matrix Basics & Notation	Pg 1
3. Types of Matrices (Table)	Pg 2	4. Matrix Operations	Pg 3
5. Rank of a Matrix	Pg 4	6. Echelon & Normal Form	Pg 5
7. Cauchy–Binet Formula	Pg 5	8. Inverse of a Matrix	Pg 6
9. System of Linear Equations	Pg 7	10. Rouché–Capelli Theorem	Pg 7
11. Gauss Elimination Method	Pg 8	12. Iterative Methods (Jacobi & G-S)	Pg 8
13. Flash Cards (25 Q&A)	Pg 9	14. Exam Tips & Tricks	Pg 10
15. Quick Revision Sheet (All Formulas)	Pg 11		

📌 1. COURSE OBJECTIVES

What You Learn:

- Understand and classify matrices
- Perform matrix operations (add, multiply, transpose)
- Find rank using echelon & normal forms
- Solve systems of linear equations ($AX = B$)
- Apply Gauss Elimination & iterative methods
- Determine consistency using Rouché–Capelli theorem

Real-World Relevance:

- Circuit analysis (KVL/KCL mesh equations)
- Structural engineering (load distribution)
- Computer graphics (transformations)
- Machine learning (data representation)
- Control systems (state-space models)
- Network flows and optimization



2. MATRIX BASICS — QUICK RECALL

Definition:

A *rectangular array* of numbers (elements) arranged in rows and columns, enclosed in brackets [] or ().

Order / Size:

Matrix of order $m \times n$ has **m rows** and **n columns**. Total elements = $m \times n$.

Notation:

Matrix: $A = [a_{ij}]$
where i = row index (1 to m),
 j = column index (1 to n).

GENERAL FORM — $M \times N$ MATRIX

$$A = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$



3. TYPES OF MATRICES

Type	Definition / Condition	Example / Key Property
Row Matrix	Only 1 row; order $1 \times n$	$A = [1 \ 2 \ 3] \rightarrow 1 \times 3$
Column Matrix	Only 1 column; order $m \times 1$	$B = [4; 5; 6] \rightarrow 3 \times 1$
Square Matrix	Number of rows = number of columns; $m = n$	Order $n \times n$; has a principal diagonal
Zero (Null) Matrix	All elements are zero; $a_{ij} = 0$ for all i, j	$O = [0 \ 0; 0 \ 0]$; $A + O = A$
Identity Matrix	Square; diagonal = 1, off-diagonal = 0; $a_{ij} = \delta_{ij}$	$I_2 = [1 \ 0; 0 \ 1]$; $AI = IA = A$
Diagonal Matrix	Square; all off-diagonal elements = 0 ($a_{ij} = 0, i \neq j$)	$D = [3 \ 0; 0 \ 5]$
Scalar Matrix	Diagonal matrix where all diagonal elements are equal ($= k$)	$S = [k \ 0; 0 \ k] = k \cdot I$
Symmetric Matrix	Square; $a_{ij} = a_{ji}$ for all $i, j \rightarrow A = A^T$	$[1 \ 2; 2 \ 5]$; $A^T = A$ always
Skew-Symmetric	Square; $a_{ij} = -a_{ji} \rightarrow A^T = -A$; all diagonal = 0	$[0 \ 3; -3 \ 0]$; $a_{ii} = 0$
Upper Triangular	All elements below main diagonal = 0 ($a_{ij} = 0, i > j$)	$[1 \ 2 \ 3; 0 \ 4 \ 5; 0 \ 0 \ 6]$
Lower Triangular	All elements above main diagonal = 0 ($a_{ij} = 0, i < j$)	$[1 \ 0 \ 0; 2 \ 3 \ 0; 4 \ 5 \ 6]$
Orthogonal Matrix	$A \cdot A^T = A^T \cdot A = I$, so $A^{-1} = A^T$	Preserves distances & angles; $ A = \pm 1$
Singular Matrix	$ A = 0$; inverse does not exist	Rank $< n$; rows/cols linearly dependent
Non-Singular Matrix	$ A \neq 0$; inverse exists	Full rank; rows/cols linearly independent
Idempotent Matrix	$A^2 = A$	Eigenvalues are 0 or 1

⚡ **Key Relations:** Every square matrix A can be written as: $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$ = (Symmetric part) + (Skew-symmetric part)

 4. MATRIX OPERATIONS

Operation	Formula / Rule	Condition	Key Property
Addition	$(A+B)_{ij} = a_{ij} + b_{ij}$	A and B must be same order (m×n)	Commutative: $A+B = B+A$; Associative: $(A+B)+C = A+(B+C)$
Subtraction	$(A-B)_{ij} = a_{ij} - b_{ij}$	Same order (m×n)	$A - B \neq B - A$ (NOT commutative)
Scalar Mult.	$(kA)_{ij} = k \cdot a_{ij}$	k is a scalar; any matrix	$k(A+B) = kA + kB$; $(k+I)A = kA + IA$
Multiplication	$(AB)_{ij} = \sum a_{ik} \cdot b_{kj}$	$\text{Cols}(A) = \text{Rows}(B)$: $A(m \times p)$, $B(p \times n) \rightarrow AB(m \times n)$	NOT commutative ($AB \neq BA$ generally); Associative: $A(BC) = (AB)C$
Transpose	$(A^T)_{ij} = A_{ji}$	$A(m \times n) \rightarrow A^T(n \times m)$	$(A^T)^T = A$; $(AB)^T = B^T A^T$; $(A+B)^T = A^T + B^T$

 Multiplication Quick Example (2×2):

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$

$AB = \begin{bmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{bmatrix}$

Row of A · Column of B → each element of AB

 Important Transpose Rules:

- $(AB)^T = B^T A^T$ **REVERSE ORDER**
- $(ABC)^T = C^T B^T A^T$
- $(kA)^T = kA^T$
- If A is symmetric: $A^T = A$
- If A is skew-sym: $A^T = -A$
- $(A^{-1})^T = (A^T)^{-1}$

 **Cancellation Laws DO NOT hold:** $AB = AC$ does NOT always mean $B = C$. | $AB = 0$ does NOT always mean $A = 0$ or $B = 0$.

 5. RANK OF A MATRIX

Definition: Rank of matrix A = *maximum number of linearly independent rows (or columns)* in A.

Denoted: $\rho(A)$ or $\text{rank}(A)$

Via Echelon Form:

Rank = Number of non-zero rows in the row echelon form of A.

Via Normal Form:

A is reduced to normal (canonical) form:

$N = [I_r \ 0 \ ; \ 0 \ 0]$

where $r = \text{rank}(A) = \text{order of identity sub-matrix}$.

KEY RANK PROPERTIES

- $0 \leq \text{rank}(A) \leq \min(m, n)$
- $\text{rank}(A) = \text{rank}(A^T)$
- $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$
- $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$
- $\text{rank}(kA) = \text{rank}(A), k \neq 0$
- For $n \times n$ matrix: $\text{rank}(A) = n \Leftrightarrow |A| \neq 0$
- $\text{rank}(O) = 0$ (zero matrix)
- $\text{rank}(I_n) = n$
- $\text{rank}(A) = \text{rank}(A^T A) = \text{rank}(A A^T)$

6. ECHELON FORM & NORMAL FORM

Row Echelon Form (REF) — Conditions:

A MATRIX IS IN REF IF:

- ⓐ All zero rows are at the BOTTOM
- ⓑ Leading entry (pivot) of each non-zero row is to the RIGHT of the pivot above it
- ⓒ All entries BELOW each pivot = 0

Reduced Row Echelon Form (RREF):

ADDITIONAL CONDITION FOR RREF:

- ⓐ Each pivot = 1
- ⓑ All entries ABOVE each pivot = 0 (only non-zero element in its column)

Row Reduction — Elementary Operations:

Three Elementary Row Operations (ERO):

- $R_i \leftrightarrow R_j$ — Interchange two rows
- $R_i \rightarrow k \cdot R_i$ — Multiply row by non-zero scalar k
- $R_i \rightarrow R_i + k \cdot R_j$ — Add multiple of one row to another

Normal Form (Canonical Form):

Any $m \times n$ matrix A of rank r is equivalent to:

$$N = [I_r \ 0 \ ; \ 0 \ 0]_{m \times n}$$

Use both row AND column operations to achieve this.

- ⚡ **Shortcut:** Rank = r = size of largest non-zero sub-matrix determinant (minor) $\neq 0$.

⚠ **Key Shortcut — Rank from Determinant:** If all $r \times r$ minors $\neq 0$, but all $(r+1) \times (r+1)$ minors = 0, then rank = r .

7. CAUCHY-BINET FORMULA

STATEMENT

If A is $m \times n$ and B is $n \times m$ (with $m \leq n$), then: $\det(AB) = \sum \det(A[S]) \cdot \det(B[S])$ where the sum is over all m -element subsets S of $\{1, 2, \dots, n\}$, and $A[S]$ is the $m \times m$ sub-matrix of A using columns in S , and $B[S]$ is the $m \times m$ sub-matrix of B using rows in S . Special case ($m = n = \text{square}$): $\det(AB) = \det(A) \cdot \det(B)$

When is it Used?

- Computing $\det(AB)$ without multiplying first
- Proving rank inequalities
- In advanced linear algebra proofs
- **Most common exam use:** Square case $\rightarrow |AB| = |A| \cdot |B|$

Related Properties:

- $|A^n| = |A|^n$
- $|kA| = k^n |A|$ for $n \times n$ matrix
- $|A^T| = |A|$
- $|A^{-1}| = 1/|A|$



8. INVERSE OF A MATRIX

MAIN FORMULA

$A^{-1} = \text{adj}(A) / |A|$ where: $\text{adj}(A)$ = Transpose of Cofactor Matrix = $[C_{ij}]^T$ Cofactor $C_{ij} = (-1)^{(i+j)} \cdot M_{ij}$ M_{ij} = Minor (det of matrix after deleting row i , col j)

Existence Condition:

A^{-1} exists $\leftrightarrow |A| \neq 0 \leftrightarrow A$ is non-singular

VERIFICATION

$A \cdot A^{-1} = A^{-1} \cdot A = I$ $(A^{-1})^{-1} = A$ $(AB)^{-1} = B^{-1} \cdot A^{-1}$
← Reverse order! $(A^T)^{-1} = (A^{-1})^T$ $(kA)^{-1} = (1/k) A^{-1}$

For 2x2 Matrix — Quick Formula:

$$A = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

$|A| = ad - bc$ $A^{-1} = (1/(ad-bc)) \cdot \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ ← Swap diagonal, negate off-diagonal

Gauss-Jordan Method (for inverse):

- 1 Form augmented matrix $[A \mid I]$
- 2 Apply row operations to left side
- 3 Reduce left side to identity I
- 4 Right side becomes A^{-1}

$[A \mid I] \rightarrow \text{Row operations} \rightarrow [I \mid A^{-1}]$

adj(A) Properties:

- $A \cdot \text{adj}(A) = |A| \cdot I$
- $|\text{adj}(A)| = |A|^{(n-1)}$ for $n \times n$
- $\text{adj}(AB) = \text{adj}(B) \cdot \text{adj}(A)$



9. SYSTEM OF LINEAR EQUATIONS

MATRIX FORM

$AX = B$ where: A = Coefficient matrix ($m \times n$) X = Variable (unknown) matrix ($n \times 1$) B = Constant matrix ($m \times 1$) $[A \mid B]$ = Augmented matrix ($m \times (n+1)$)

EXPANDED FORM (3 EQUATIONS, 3 UNKNOWNNS)

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned}$$

Consistency Conditions:

Condition	Type of System	Solutions
$\rho(A) = \rho([A B]) = n$	Consistent	Unique Solution
$\rho(A) = \rho([A B]) < n$	Consistent	Infinite Solutions ($n - r$ free vars)
$\rho(A) \neq \rho([A B])$	Inconsistent	No Solution

Homogeneous System ($B = 0$, $AX = 0$):

- Always consistent ($X = 0$ is trivial solution)
- Non-trivial solutions exist $\leftrightarrow \rho(A) < n \leftrightarrow |A| = 0$
- Number of free variables = $n - \rho(A)$



10. ROUCHÉ-CAPELLI THEOREM

Formal Statement: A system of linear equations $AX = B$ is *consistent* (has at least one solution) if and only if the rank of the coefficient matrix A equals the rank of the augmented matrix $[A|B]$.

$$\text{System is consistent} \leftrightarrow \rho(A) = \rho([A|B])$$



Unique Solution

$$\rho(A) = \rho([A|B]) = n$$

(n = number of unknowns)

System has exactly 1 solution



Infinite Solutions

$$\rho(A) = \rho([A|B]) < n$$

Infinite solutions; ($n-r$) free variables where r = rank



No Solution

$$\rho(A) \neq \rho([A|B])$$

$\rho([A|B]) = \rho(A) + 1$; System is inconsistent



Simple Words: If adding the constants column (B) to matrix A increases the rank, then those constants are "incompatible" with the equations $\rightarrow$ No solution exists. If rank stays same, the system is solvable.



11. GAUSS ELIMINATION METHOD

Steps Summary:

- 1 Write augmented matrix $[A \mid B]$
- 2 Use EROs to reduce to Upper Triangular form (forward elimination)
- 3 Back-substitute starting from last equation upward
- 4 Obtain values of $x_1, x_2, \dots, x_n$

Key Idea: Systematically eliminate variables from equations to create a triangular system, then solve by back-substitution.

FORWARD ELIMINATION — PIVOT OPERATION

To eliminate variable x_k from row i ($i > k$): $R_i \rightarrow R_i - (a_{ik} / a_{kk}) \cdot R_k$ where a_{kk} = pivot element (must be non-zero; swap rows if needed)

Gauss-Jordan Extension: Continue reduction all the way to RREF (also eliminate above pivots) $\rightarrow$ Direct solution without back-substitution.

Partial Pivoting: If pivot = 0, swap with the row below having the largest absolute value in that column (numerical stability).



Setup: For system $AX = B$, rearrange each equation to express each variable in terms of others. Starting with initial guess (usually $X^0 = 0$), iterate until convergence.

Jacobi Method

ITERATION FORMULA (FOR 3×3 SYSTEM)

$$\begin{aligned}x_1^{(k+1)} &= (b_1 - a_{12}x_2^{(k)} - a_{13}x_3^{(k)}) / a_{11} \\x_2^{(k+1)} &= (b_2 - a_{21}x_1^{(k)} - a_{23}x_3^{(k)}) / a_{22} \\x_3^{(k+1)} &= (b_3 - a_{31}x_1^{(k)} - a_{32}x_2^{(k)}) / a_{33}\end{aligned}$$

Key Rule: Use OLD values (iteration k) of ALL variables to compute ALL new values simultaneously.

Matrix Form: $X^{(k+1)} = D^{-1}(B - (L+U)X^{(k)})$
where D = diagonal, L = lower, U = upper parts of A .

Gauss-Seidel Method

ITERATION FORMULA (FOR 3×3 SYSTEM)

$$\begin{aligned}x_1^{(k+1)} &= (b_1 - a_{12}x_2^{(k)} - a_{13}x_3^{(k)}) / a_{11} \\x_2^{(k+1)} &= (b_2 - a_{21}x_1^{(k+1)} - a_{23}x_3^{(k)}) / a_{22} \uparrow \text{NEW value used!} \\x_3^{(k+1)} &= (b_3 - a_{31}x_1^{(k+1)} - a_{32}x_2^{(k+1)}) / a_{33} \uparrow \text{NEW values used!}\end{aligned}$$

Key Rule: Use LATEST (most updated) values as soon as they are available.

Matrix Form: $X^{(k+1)} = (D+L)^{-1}(B - UX^{(k)})$

Feature	Jacobi Method	Gauss-Seidel Method
Values used	All OLD values from iteration k	Latest UPDATED values (mixed)
Update pattern	Simultaneous (all at once)	Sequential (one by one)
Convergence speed	Slower	Faster (about 2× Jacobi)
Can be parallelized?	Yes (easier)	Harder (sequential dependency)
Memory needed	Two sets of values	One set (in-place update)

⚡ **Convergence Condition (Diagonal Dominance):**
The method is guaranteed to converge if A is *strictly diagonally dominant*:

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}| \quad \text{for every row } i$$

i.e., Each diagonal element's magnitude exceeds the sum of all other elements' magnitudes in that row.

Q1: What is the rank of a matrix?

A: Maximum number of linearly independent rows (or columns). Also = order of the largest non-zero minor.

Q3: When does a matrix NOT have an inverse?

A: When $|A| = 0$ (singular matrix). Inverse: $A^{-1} = \text{adj}(A)/|A|$, needs $|A| \neq 0$.

Q5: State Rouché–Capelli Theorem.

A: $AX = B$ is consistent $\leftrightarrow \rho(A) = \rho([A|B])$. If $\rho(A) \neq \rho([A|B])$, no solution exists.

Q7: What is the key difference between Jacobi and Gauss-Seidel?

A: Jacobi uses old values for all unknowns; Gauss-Seidel uses the latest updated values immediately.

Q9: What is the diagonal of a skew-symmetric matrix?

A: All diagonal elements = 0. Since $a_{ii} = -a_{ii} \rightarrow 2a_{ii} = 0 \rightarrow a_{ii} = 0$.

Q11: What is $(AB)^{-1}$?

A: $(AB)^{-1} = B^{-1} \cdot A^{-1}$. Order is reversed! (Like removing shoes: put them on A then B, remove B first.)

Q13: What is an idempotent matrix?

A: $A^2 = A$. Its eigenvalues are 0 or 1 only. Example: any projection matrix.

Q15: What is the convergence condition for iterative methods?

A: Strict diagonal dominance: $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ for all rows i .

Q17: If $\rho(A) = \rho([A|B]) = r < n$, how many free variables?

A: $n - r$ free variables, giving infinitely many solutions parameterized by $n - r$ parameters.

Q19: What is $\text{rank}(A)$ if A is a square non-singular $n \times n$ matrix?

A: $\text{rank}(A) = n$. Full rank $\leftrightarrow$ non-singular $\leftrightarrow |A| \neq 0 \leftrightarrow$ inverse exists.

Q21: What is $|kA|$ for an $n \times n$ matrix?

A: $|kA| = k^n \cdot |A|$. Each row gets multiplied by k , so determinant is multiplied by k^n .

Q23: Every square matrix can be written as sum of what?

A: Sum of a symmetric matrix and a skew-symmetric matrix: $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$.

Q25: What is the Cauchy-Binet formula for square matrices?

A: $\det(AB) = \det(A) \cdot \det(B)$. Determinant of a product = product of determinants.

Q2: What is the condition for $AX = B$ to have a unique solution?

A: $\rho(A) = \rho([A|B]) = n$, where n = number of unknowns.

Q4: What is a symmetric matrix?

A: A square matrix where $A = A^T$, i.e., $a_{ij} = a_{ji}$ for all i, j .

Q6: What is the normal form of a matrix of rank r ?

A: $N = [I_r, 0; 0, 0]$, where I_r is the $r \times r$ identity matrix, using row and column operations.

Q8: What is the rank of an identity matrix I_n ?

A: $\text{rank}(I_n) = n$. All rows are linearly independent.

Q10: Formula for inverse of 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$?

A: $A^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} / (ad - bc)$. Swap diagonal, negate off-diagonal, divide by determinant.

Q12: What does homogeneous system $AX = 0$ always have?

A: Always has trivial solution $X = 0$. Non-trivial solutions exist only if $|A| = 0$ ($\text{rank} < n$).

Q14: What is $(A^T)^T$?

A: $(A^T)^T = A$. Double transpose returns the original matrix.

Q16: What are the three types of elementary row operations?

A: (1) Swap rows $R_i \leftrightarrow R_j$, (2) Scale: $R_i \rightarrow kR_i$, (3) Add multiple: $R_i \rightarrow R_i + kR_j$.

Q18: What is $|\text{adj}(A)|$ for an $n \times n$ matrix A ?

A: $|\text{adj}(A)| = |A|^{n-1}$. For 3×3 : $|\text{adj}(A)| = |A|^2$.

Q20: How many solutions does $AX = B$ have if A is non-singular?

A: Exactly one unique solution: $X = A^{-1}B$ (multiply both sides by A^{-1} on the left).

Q22: What is the Gauss-Jordan method used for?

A: Finding inverse of a matrix: augment A with I , reduce $[A|I]$ to $[I|A^{-1}]$ using row operations.

Q24: What is $\text{rank}(A) + \text{nullity}(A)$?

A: $\text{rank}(A) + \text{nullity}(A) = n$ (number of columns). This is the Rank-Nullity Theorem.

✖ Common Mistakes Students Make:

Mistake 1: Forgetting to check if $|A| \neq 0$ before computing A^{-1} . Always verify first!

Mistake 2: Writing $(AB)^{-1} = A^{-1}B^{-1}$. WRONG! Correct: $(AB)^{-1} = B^{-1}A^{-1}$ (reversed order).

Mistake 3: Confusing $\rho([A|B])$ and $\rho(A)$. Must compute BOTH ranks separately.

Mistake 4: Using Jacobi with new values (that's Gauss-Seidel). Keep them separate!

Mistake 5: Not arranging equations for diagonal dominance before applying iterative methods.

Mistake 6: Concluding "no solution" when $\text{rank}(A) = \text{rank}([A|B]) < n$. It's ∞ solutions, not no solution!

✔ Time-Saving Tricks:

⚡ For 2×2 inverse, use direct formula: swap diagonal, negate off-diagonal, divide by det. Saves minutes.

⚡ Rank of echelon form = count non-zero rows. Extremely quick to find.

⚡ For diagonal matrix, inverse = just invert each diagonal element. $|D|$ = product of diagonal elements.

⚡ Check diagonal dominance FIRST before iterative method. If not dominant, rearrange equations.

⚡ If $|A| = 0$, instantly conclude $\text{rank} < n$, homogeneous system has non-trivial solution, and inverse doesn't exist.

⚡ Gauss-Seidel converges $\sim 2\times$ faster than Jacobi — always prefer Gauss-Seidel if asked which is better.

🔗 Method Selection Guide:

Situation	Use This
Find rank & check consistency	Echelon Form
Solve small system (exact)	Gauss Elimination
Find inverse (systematic)	Gauss-Jordan
Large system (approximate)	Gauss-Seidel
Check consistency only	Rouché-Capelli

MATRIX BASICS

Order: $m \times n$ (m rows, n cols)

Elements: a_{ij} (row i , col j)

Square: $m = n$

Transpose: $(A^T)_{ij} = a_{ji}$

Sym: $A^T = A$

Skew-sym: $A^T = -A$

$A = \frac{1}{2}(A+A^T) + \frac{1}{2}(A-A^T)$

RANK

$\rho(O) = 0$

$\rho(I_n) = n$

$0 \leq \rho(A) \leq \min(m, n)$

$\rho(A) = \rho(A^T)$

$\rho(kA) = \rho(A), k \neq 0$

Normal form: $[I_r \ 0; \ 0 \ 0]$

$|A| \neq 0 \leftrightarrow \rho = n$ (full rank)

ROUCHÉ-CAPELLI

$AX = B$ consistent iff

$\rho(A) = \rho([A|B])$

Unique sol: $\rho(A) = \rho([A|B]) = n$

∞ sols: $\rho(A) = \rho([A|B]) < n$

No sol: $\rho(A) \neq \rho([A|B])$

Free vars: $n - \rho(A)$

Homogeneous: always consistent

JACOBI METHOD

Use **OLD** values (k) for all

$x_1^{(k+1)} = (b_1 - a_{12}x_2^{(k)} - a_{13}x_3^{(k)}) / a_{11}$

$x_2^{(k+1)} = (b_2 - a_{21}x_1^{(k)} - a_{23}x_3^{(k)}) / a_{22}$

$x_3^{(k+1)} = (b_3 - a_{31}x_1^{(k)} - a_{32}x_2^{(k)}) / a_{33}$

Start: $x^0 = (0, 0, 0)$

Converge: diagonal dominance

DETERMINANT RULES

$|A^T| = |A|$

$|AB| = |A| \cdot |B|$

$|kA| = k^n |A|$ ($n \times n$)

$|A^{-1}| = 1/|A|$

$|A^n| = |A|^n$

Swap rows: sign changes

Singular: $|A| = 0$

OPERATIONS

Add: $(A+B)_{ij} = a_{ij} + b_{ij}$

Mult: $(AB)_{ij} = \sum a_{ik}b_{kj}$

Cond: Cols(A) = Rows(B)

NOT commutative: $AB \neq BA$

$(AB)^T = B^T A^T$

$(AB)^{-1} = B^{-1} A^{-1}$

$|AB| = |A| \cdot |B|$

INVERSE

$A^{-1} = \text{adj}(A) / |A|$

Exists iff $|A| \neq 0$

2×2 : $A^{-1} = [d \ -b; -c \ a] / (ad-bc)$

$|\text{adj}(A)| = |A|^{(n-1)}$

$A \cdot \text{adj}(A) = |A| \cdot I$

$(A^{-1})^T = (A^T)^{-1}$

G-J: $[A|I] \rightarrow [I|A^{-1}]$

GAUSS ELIMINATION

Form: $[A \ | \ B]$

Pivot op: $R_i \rightarrow R_i - (a_{ik}/a_{kk})R_k$

Goal: Upper triangular form

Then: **Back-substitution**

G-J: Continue to RREF

Partial pivot if $a_{kk} = 0$

Swap with max|element| row

GAUSS-SEIDEL METHOD

Use **LATEST** values immediately

$x_1^{(k+1)} = (b_1 - a_{12}x_2^{(k)} - a_{13}x_3^{(k)}) / a_{11}$

$x_2^{(k+1)} = (b_2 - a_{21}x_1^{(k+1)} - a_{23}x_3^{(k)}) / a_{22}$

$x_3^{(k+1)} = (b_3 - a_{31}x_1^{(k+1)} - a_{32}x_2^{(k+1)}) / a_{33}$

~2x faster than Jacobi

Converge: $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$

KEY IDENTITIES

$A + O = A$

$AI = IA = A$

$A \cdot A^{-1} = I$

$A^n \cdot A^m = A^{(n+m)}$

rank + nullity = n

Orthogonal: $A^{-1} = A^T$

Idempotent: $A^2 = A$

LAST-MINUTE MUST-REMEMBER (TOP 10 FACTS)

① $A^{-1} = \text{adj}(A) / |A| \rightarrow$ needs $|A| \neq 0$

② $\text{rank}(A) = \#$ non-zero rows in REF

③ $\rho(A) = \rho([A|B]) = n \rightarrow$ Unique solution

④ Diagonal dominance $\rightarrow$ convergence guaranteed

⑤ Normal form = $[I_r \ 0; \ 0 \ 0]$ where $r = \text{rank}$

⑥ $(AB)^{-1} = B^{-1}A^{-1} \rightarrow$ reversed order

⑦ $\rho(A) \neq \rho([A|B]) \rightarrow$ No solution

⑧ Gauss-Seidel converges faster than Jacobi

⑨ Sym: $A = A^T$ | Skew-sym: $A^T = -A$, $\text{diag} = 0$

⑩ $AX=0$ always consistent; nontrivial $\leftrightarrow |A|=0$